INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – I , First Semester Examination, August- 2025

Subject Code: BCACC102

Subject Name: Digital Electronics

Full Marks: 30 Duration: 1 Hour

Part - A

Answer any 5 questions out of 6

Each question carries 2 marks (2 × 5)

1. Convert the decimal number $(444.456)_{10}$ to octal number. 2
2. Subtract $(1110)_2$ and $(1101)_2$ using 2's complement method. 2
3. Implement an OR gate using NAND gate. 2
4. Prove A + AB = A 2
5. What is a forbidden state in SR flipflop? 2
6. Differentiate between latch and flipflop. 2

Part - B

Answer any 2 questions out of 3

Each question carries 5 marks (5 × 2)

7. If $A\bar{B} + \bar{A}B = C$, show that $A\bar{C} + \bar{A}C = B$ 5

8. What are non-weighted codes? List any two non-weighted codes and explain why they are called non-weighted. Also, give a 4-bit example for each. 5

9. What is meant by a functionally complete set of logic gates? Prove that the NAND gate alone is functionally complete by implementing the basic gates (NOT, AND, OR) using only NAND gates. 5

Part - C

Answer any 1 question out of 2

Each question carries 10 marks (10 × 1)

10. Describe the working of a NAND based SR flipflop with the help of a circuit diagram, characteristics table and excitation table. 10

11. Realise the function f = B(A+CD) +AC' as (i) multilevel NAND-NAND gate network and (ii) multilevel NOR-NOR network. 10

--End--